M T W T F S DATE:

SProb ICL=

Uncertainity Due To Randomness:

x_1 x_2

marks = f(Std Hrs, Travel Hrs)

$y = a_0 + ax_1 + ax_2$

$y = 1 + 2x_1 + 0.5x_2$

↓

ALL are Controllable.

- But if accident, temp: is uncontrollable
  ↳ these are due to randomness.

Prob Theory: Depends on event that happen or occur in Future)

Why Prob:

- Model the Randomness.
- Generalize.
- Probability & Inferent

Popul

Real Pop (Unit physically exist)

Hypothetical (Unit doesn't exist)

Generalize (Infere)

Pop ← sample

↓ ↓

Deserip Discrip

SPROB 1 CL:

Hypothetical Population
- outcome of an experiment
- Exp:-
  - An activity that produce some results

Result

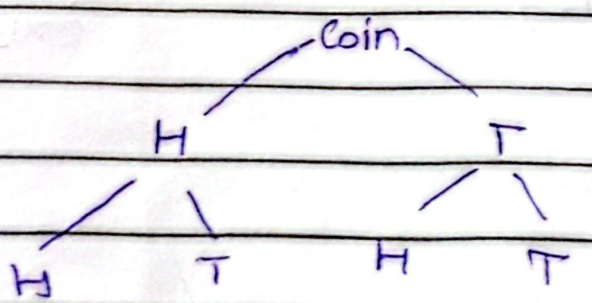
Predictable → If Predictable then Deterministic

Unpredictable → If unpredicable then Random. (e.g. chipkilli)

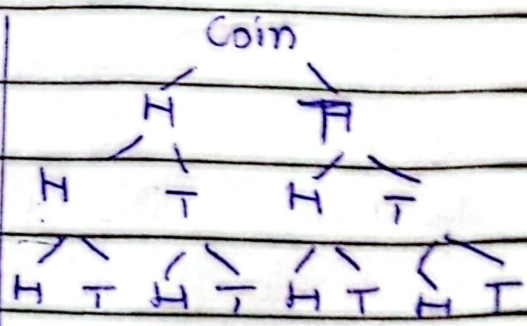
even though repeated Large number of times under same condition.

Trial: single Perform of an experiment.

Sample Space: set of ALL Possible Outcomes



HH, HT, TH, TT



~~HHH~~, ~~HHT~~, ~~HTH~~, ~~HTT~~
~~THH~~, ~~THT~~, ~~TTH~~, ~~TTT~~

M T W T F S DATE:

MTH 262 → Make Sample Space

S = {A, A⁻, B, B⁻, B⁺, C, C⁻, C⁺, D, D⁺, F}

MTH262

	A	B	C	D	F
A	(A,A)	(A,B)	(A,C)		
B					
C					
D					
F					

CSC210

Event: Subset of sample space.

Event: S = {HH, HT, TH, TT} → Sure Event

Compound Event:
{HH}, {HT}, {HH, TH}, {HH, TT}
{HH, HT, TH}, {HH, TH, TT}, {HT, TH}
{HT, TT}, {HT, TH, TT}, {TH, TT}

{HH}, {HT}, {TH}, {TT}, ∅ → NULL Event

Simple Event

Mutually Exclusive.

A ∩ B = ∅ → only one event can occur at a time → Like LG will be either A, A etc

Exhaustive Event: A ∪ B = ∅ if take Union of event and they comes as sample space

Equally Likely: both event has equal chance

M T W T F S DATE: 8/10/25

Probab

objective
- (Measure chance by some well define method)
- Prior knowLeadge matters.

Subjective:
- (some one personally decide about something)

classical definition.

Relative Freq

Axiomatic definition

$P(A) = \frac{n(A)}{n(S)}$

P(event) = # favourable outcomes

possible outcomes.

Example:

Two Fair coins are tossed. what is prob of getting one heads & one tails?

S = {HH, HT, TH, TT}

P (one head & one tail)

Let A : one head & one tail

A = {HT, TH}

$$P(A) = \frac{n(A)}{n(S)} = \frac{2}{4} = 0.5 = 50\%$$

If we have 3. (Not Possible)

~~$P(A) = \frac{4}{8} = 0.5 = 50\%$~~

If we have 3 (Head, Tail, Head)

$$P(A) = \frac{1}{8} = 0.8 = 8\%$$

Prob of Event:

Example:

A card is drawn from an ordinary deck. Find these probabilities

a) of getting Jack

Let event J: → Must Define. $P(J) = \frac{n(J)}{n(S)}$

$$P(J) = \frac{(4)(48)}{\binom{52}{1}}$$

b) of getting 6 of club

Let event C:

$$P(C) = \frac{1}{52} = 0.08$$

c) of getting 3 or a Diamond.

DATE:

order & unorder.

N = A, B, C

n = 2

1st	2nd	
A	B	
A	C	
B	C	No meaningful order so (combination)
B	A	
C	A	

Now suppose (Permutation).

we want to select two student First will get additional 10 marks then we use Lottery method and now it will be meaningful.

$\frac{52!}{0(52-0)!} = 1$

Example:

committee of 3 person is to be selected randomly from a group of 5 and 3 woma

or

N = 8

selects = n = 3

$^nC_r = \frac{n!}{(n-r)!\,r!}$

M T W T F S

DATE:

Let event: A (selecting 1 men 8 women)

$$P(A) = \frac{\binom{5}{1}\binom{3}{2}}{\binom{8}{3}} = \frac{\frac{5!}{1!(5-1)!}\cdot\frac{3!}{2!(3-2)!}}{\frac{8!}{3!(8-3)!}} = \frac{(5)(3)}{56} = \frac{15}{56} = 0.27 \downarrow$$

27%

b) –

Let event: B (selecting 3 women)

$$P(B) = \frac{\binom{5}{0}\binom{3}{3}}{\binom{8}{3}} = \frac{\left(\frac{5!}{0!(5-0)!}\right)\left(\frac{3!}{3!(3-3)!}\right)}{\left(\frac{8!}{3!(8-3)!}\right)}$$

$$P(E) = \frac{(1)(1)}{56} = 0.02 = 2\%$$

Given Data:

Select = 3 Men = 5

Total = 8 Women = 3

$$n(S) = \binom{8}{3} = \frac{8!}{3!(8-3)!} = 56$$

$$P(A) = \frac{n(A)}{n(S)} = ($$

$$P(B) = \frac{n(B)}{n(S)}$$

M T W T F S DATE:

Emperical or Relative Frequency Approach

LG	f	RF
A	8	0.22
B	9	0.25
C	10	0.28
D	5	0.14
F	3	0.08
	35	1

$$P(E) = \frac{f}{n} = \frac{\text{Freq. For Class}}{\text{T. Freq. in Dist.}}$$

Find out Prob For MTH262 (of getting A)

P(A or F) & P(A and F)

Let event A: (P and F).

$P(A) = P(A) \cdot P(F)$ (For 1 Subject its not possible it mutually Exclusive)

$P(A) = (0.22)($